

AI CCTV Event Summarizer using VL-JEPA + Small Language Models

Deep Research Report — Feasibility, Novelty, and Commercialization Analysis

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Executive Summary

The proposed pipeline — self-supervised video representation → multi-module event detection → structured event store → vector search → small-LLM report generation — is **architecturally sound and directionally aligned with where the industry is actually moving in 2025–2026**, but it is **not a greenfield idea**. NVIDIA's Metropolis "Video Search and Summarization" (VSS) Blueprint, Verkada's AI-powered search/alerts/incident-summary stack, and a fast-growing academic literature on VLM-based video anomaly detection and VideoRAG have already validated almost every individual component described in the prompt. The genuinely novel element is a **very recent (Dec 2025/arXiv 2512.10942) research-stage model called VL-JEPA**, a non-autoregressive, embedding-predicting vision-language architecture — distinct from Meta's V-JEPA/V-JEPA 2, which are vision-only world models with no native language head. No commercial product and no published paper (as of this research) combines VL-JEPA specifically with a small on-device LLM for structured CCTV incident reporting. That specific combination is your strongest, most defensible novelty claim — but it should be framed as "first application/adaptation of a two-month-old architecture to a new domain," not as "no one has done automated CCTV summarization before," because many people have.

Bottom line: feasible as a capstone if scoped down from the full 20+-event, multi-module vision (which is really a product roadmap, not a thesis), genuinely novel at the *combination* level rather than the component level, and publishable as a workshop/applied-track paper with the right experiments — but it needs one or two sharp research contributions layered on top rather than being pitched as "build the whole platform."

1. Existing Commercial Systems

Company	Core tech	Natural-language reports?	Semantic/NL search?	Multimodal / self-supervised?	Notes
BriefCam (now part of Milestone Systems)	Deep learning object/attribute classifiers + patented VIDEO SYNOPSIS® (overlays multi-timestamp events on one frame)	No native LLM report writer; strong structured dashboards/exports instead	Filter-based search on attributes (not fully open natural language)	Multimodal in the sense of face/LPR/behavior fusion, not self-supervised video embeddings	Industry incumbent for post-event investigation; being absorbed into Milestone's XProtect ecosystem as of early 2026
Verkada	Cloud + on-camera CV, Ambarella CV7x edge chips	Yes — as of March 2026, one-click AI-generated incident summaries auto-describe archived footage in incident reports; fighting/fall detection auto-triggers reports	Yes — freeform natural-language "AI-powered Search" ("person wearing red jacket near loading dock"), plus AI-powered Alerts from NL queries, plus NL audit-log search (May 2026)	Uses appearance-embedding + facial-analytics fusion; not disclosed as self-supervised JEPA-style pretraining	Closest commercial analog to the proposed system's "NL search + auto-report" vision — already shipping
NVIDIA Metropolis / VSS Blueprint	VLMs (VILA, Nemotron/Nemotron-3-Nano-Omni) + LLMs + RAG (NeMo Retriever)	Yes — "summarize / alert / report / review" are native agent skills (VSS 3.x, mid-2026)	Yes — full RAG-based NL search over live and archived video, multi-video reports	Yes — VLM+LLM+RAG pipeline is essentially the architecture proposed in this prompt, at reference-implementation level, open-sourced on GitHub	This is the single most important prior-art system to disclose. It is not a finished vertical product but a <i>developer blueprint</i> — companies like Linker Vision (smart cities) and VAST (NHL footage) build on top of it. A capstone that says "I built VSS-inspired pipeline with VL-JEPA instead of VILA" is honest and still defensible
Axis Communications	Edge deep-learning cameras (ACAP platform), partner ecosystem (hosts BriefCam, others)	No native LLM reporting	Basic attribute search via partners	Edge AI, not self-supervised	Mostly a hardware/platform play; analytics comes from ecosystem partners

Company	Core tech	Natural-language reports?	Semantic/NL search?	Multimodal / self-supervised?	Notes
Hikvision / Dahua	In-house deep learning ("AI Cloud"/"HikCentral", "DeepSense"), strong in China/APAC/government	Emerging generative-AI features (2026 product cycles) but largely detection/classification-centric	Attribute-based search (face, plate, human/vehicle)	Primarily supervised detection models	Massive scale, cost-competitive, but weaker in the "semantic reasoning/report" layer historically; regulatory scrutiny in Western markets
Milestone Systems (XProtect)	VMS platform; now owns BriefCam	No native LLM summarizer	Via BriefCam module	Same as BriefCam	Distribution/VMS backbone, not itself a reasoning engine
Genetec (Security Center)	VMS + access control + LPR + Genetec-native analytics	No native LLM summarizer disclosed	Attribute/appearance search	Supervised CV	Strong in unified security ops, weaker on generative/semantic layer
AWS (Rekognition Video)	CNN-based label/activity/face detection, custom labels	No native LLM report generation (would need Bedrock integration by the customer)	Basic label/attribute search	Not self-supervised out of the box	Building block, not an end-to-end reporting product — a capstone could legitimately position itself as "what AWS doesn't give you out of the box"
Google Cloud Video Intelligence API	Label/shot/object detection, explicit content detection	No native summarizer (would need Vertex AI + Gemini integration)	Label-based	Not self-supervised	Same positioning as AWS — infra primitive, not a finished report generator
Protex AI / viAct / Securade.ai (industrial safety niche)	CV-based unsafe-behavior detection, PPE, near-miss capture, plugs into existing CCTV	Yes — Protex AI markets a "powerful reporting engine" and a natural-language "Protex Intelligence" that builds dashboards on request; viAct generates compliance reports	Growing NL capability (2026 updates)	Privacy-preserving, on-prem processing; not JEPA-based	This is the closest industry-vertical match to your "worker/PPE/fall/machine" event list — worth benchmarking against directly since it targets exactly your industrial-safety use case, not general security

Overall takeaway: the "detect → structure → summarize in natural language → let users ask questions" pattern is already a 2025–2026 industry trend (Verkada, NVIDIA VSS, Protex AI all ship pieces of it). Your differentiator has to be *how* you do scene understanding (self-supervised VL-JEPA-style embeddings vs. supervised detectors/CLIP-style VLMs) and *where* you deploy (small, on-prem/edge LLM rather than cloud-scale VLM+LLM), not the existence of the report-generation idea itself.

2. Existing Research Papers (2022–2026)

Core architecture papers

VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language — Delong Chen et al., arXiv:2512.10942 (Dec 2025 / OpenReview Oct 2025).

- Architecture:** An x-encoder maps visual input to embedding S_V , a y-encoder maps text target to S_Y , and a predictor learns $(S_V, X_Q) \rightarrow S_Y$, where X_Q is a text query. Predicts continuous text embeddings instead of autoregressive tokens; a lightweight decoder is invoked only when human-readable text is needed ("selective decoding").
- Key contributions:** 2.85× reduction in decoding operations for streaming video; outperforms token-space VLMs of comparable size with 50% fewer trainable parameters; natively supports classification, retrieval, and VQA in one unified model.
- Relevance:** This is the actual "VL-JEPA" referenced in your prompt. It is **extremely recent** — published essentially at your project's start date — which is both an opportunity (genuine first-mover territory) and a risk (immature tooling, no production-grade checkpoints/ecosystem yet, thin community support).
- Limitations/future work:** As a brand-new architecture, long-horizon video streaming behavior, robustness on out-of-distribution surveillance-style footage, and integration with downstream LLMs for generative reporting are all unexplored in the original paper.

V-JEPA / V-JEPA 2 — Meta AI (Assran, Bardes et al.), arXiv:2506.09985, June 2025.

- Self-supervised, action-free JEPA pretrained on 1M+ hours of video; predicts masked spatiotemporal patches in latent space rather than pixels; 1.2B-parameter ViT-g encoder; 77.3 top-1 on Something-Something v2, SOTA on Epic-Kitchens-100 action anticipation; when aligned with an LLM, SOTA video-QA at 8B scale.
- Important clarification for your proposal:** V-JEPA/V-JEPA 2 is **vision-only** (a "world model" for physical dynamics/robotics), not natively vision-language. If your architecture diagram cites "VL-JEPA," make sure you're citing the correct paper (2512.10942) and not conflating it with V-JEPA 2 — reviewers and evaluators will check this.

Video anomaly detection / VLM reasoning for surveillance

- Cerberus** (arXiv 2510.16290): cascaded VLM architecture for real-time video anomaly detection.
- VadCLIP / ReVid / Holmes-VAD / SlowFastVAD / EventVAD / Flashback / MissionGNN** (2024–2025, various venues incl. AAAI, CVPR, WACV): a dense cluster of work adapting CLIP-style and generative VLMs to weakly-supervised or training-free video anomaly detection, several with explicit RAG or knowledge-graph components (MissionGNN builds a "mission-specific knowledge graph" from detected events — directly relevant to your event-database + timeline idea).
- "The Evolution of Video Anomaly Detection: A Unified Framework from DNN to MLLM"** (arXiv 2507.21649): a comprehensive 2025 survey tracing the field's move from pure DNN classifiers to multimodal LLM reasoning — useful as your primary related-work anchor.

- **AssistPDA** (arXiv 2503.21904): "online video surveillance assistant for video anomaly prediction, detection, and analysis" — conceptually very close to your full pipeline (prediction + detection + analysis in one online assistant).
- **VERA** (CVPR 2025): explainable video anomaly detection via "verbalized learning" of VLMs — relevant to your explainability/report-generation goals.

Retrieval-augmented / knowledge-graph video understanding

- **VideoRAG** (ACL Findings 2025, arXiv 2501.05874 and a second "extreme long-context" variant arXiv 2502.01549): retrieval-augmented generation directly over video corpora, selecting informative frames for retrieval+generation — structurally very close to your "event similarity search + NL search" advanced feature.
- **MissionGNN** (WACV 2025): hierarchical multimodal GNN with mission-specific knowledge-graph generation for weakly supervised anomaly recognition — a strong precedent for the "knowledge graph construction from events" improvement idea in your prompt.
- **GenDFIR** (arXiv 2409.02572): RAG + LLM for cyber-incident *timeline* analysis — not video, but structurally identical to your "Timeline Generation Engine + NL query" concept; worth citing as a cross-domain analog.

Foundational building blocks named in your stack

- **VideoMAE, SlowFast, X3D** — all mature (2021–2022), well-benchmarked action-recognition backbones; not novel on their own but reasonable engineering choices.
- **ST-GCN for pose-based fall detection** — mature (2018 origin), still actively used in 2024–2025 industrial fall-detection papers (see Section 6).
- **YOLOv11/YOLO26, ByteTrack/DeepSORT** — mature, standard detection/tracking stack; expect reviewers to treat this as "solid engineering," not "research contribution."

3. Existing Open-Source Projects

Project	Stack	Status/activity	Gap vs. your proposal
SharpAI/DeepCamera	Local VLMs (Qwen, DeepSeek, SmolVLM, LLaVA) + YOLO26, LLM-powered "agentic security camera agent," pluggable skills, hardware auto-detection (NVIDIA/AMD/Apple/Intel/CPU)	Active, actively expanding skill library	Closest OSS analog to your full vision — but it targets home/small-site security, not industrial multi-camera event timelines, PPE/machine-safety events, or vector-search-based similarity retrieval
arashsajjadi/ai-powered-video-analyzer	YOLO/D-FINE detection, BLIP captioning, Whisper transcription, PANNs audio events, Ollama LLM summarization	Active, offline-first design	Single-video analysis tool, not a live multi-camera timeline/dashboard system; no vector DB / similarity search
Ravi-Teja-konda/Surveillance_Video_Summarizer	Fine-tuned Florence-2 VLM (on SPHAR dataset) + SQLite + Gradio + OpenAI API for log Q&A	Smaller/academic-style project	Good proof that "VLM annotation → DB → LLM Q&A" works at small scale; lacks tracking, multi-camera fusion, structured severity scoring
Various "cctv-surveillance" GitHub-topic repos (YOLOv8 + ChatGPT summarization, PPE-detection repos, Shinobi CE, ZoneMinder-style NVRs)	Mixed	Many are student/hobbyist projects	Confirms the <i>general pattern</i> (YOLO + LLM narration) is already widely reproduced; a capstone repeating only this pattern without VL-JEPA-level self-supervised representation or event similarity search would not stand out from dozens of existing repos

Opportunity for improvement across all of the above: none of the surveyed open-source projects combine (a) self-supervised video-embedding backbones, (b) a vector database for cross-event visual similarity search, and (c) structured, severity-scored, multi-camera timelines feeding a small on-device LLM. That three-way combination is where your proposal can genuinely differentiate from the OSS landscape, even though each piece individually exists somewhere.

4. Novelty Assessment

- Has anyone combined VL-JEPA with surveillance event summarization?** Not found. VL-JEPA (2512.10942) is roughly six months old as an idea and has, as far as this search could determine, only been evaluated on general VQA/retrieval/classification benchmarks — not applied to surveillance, industrial safety, or CCTV timelines.
- Has anyone combined VL-JEPA with small LLMs for CCTV reports?** No. This specific combination does not appear in the literature or in any commercial product surveyed.
- Has anyone combined semantic event retrieval with NL search for surveillance?** Yes, extensively — Verkada's AI-powered Search/Alerts, NVIDIA VSS's RAG-based search, and academic VideoRAG/MissionGNN work all do this, though typically with CLIP-style contrastive embeddings or full VLMs rather than JEPA-style predictive embeddings.
- Has anyone generated automatic daily/incident surveillance reports using a comparable end-to-end architecture?** Yes at the commercial level (Verkada's one-click incident summaries, NVIDIA VSS's "report" agent skill, Protex AI's reporting engine); yes at the research level for anomaly explanation (VERA, Holmes-VAD); but not with a JEPA-style self-supervised backbone specifically, and not typically packaged as a full "daily/shift/weekly factory report" generator for industrial multi-event environments.
- Which components are already well-studied?** Object detection (YOLO family), MOT (ByteTrack/DeepSORT), action recognition (VideoMAE/SlowFast/X3D), pose-based fall detection (ST-GCN and successors), PPE detection, VLM-based anomaly detection/explanation, and RAG over video/text logs. All mature, all heavily published.

Which combinations appear new? (1) VL-JEPA as the scene-representation backbone for a surveillance pipeline; (2) VL-JEPA embeddings feeding a *small* (edge-deployable, 1–4B parameter) LLM specifically for structured, severity-scored industrial safety reporting; (3) fusing self-supervised JEPA-style embeddings (rather than CLIP/VLM embeddings) into the vector-search/event-similarity layer.

Novelty rating: **6/10**

Justification: the *system-level idea* ("understand CCTV, build a timeline, write a report, let people search it in plain English") is not novel — it is close to becoming an industry-standard pattern in 2025–2026 (Verkada, NVIDIA VSS). What can be genuinely novel is a narrower, well-defined research question about *substituting a JEPA-style self-supervised video-language backbone* for the token-generating VLMs everyone else is using, and measuring whether that changes efficiency, hallucination rate, or domain-generalization for industrial-safety event reporting specifically. That narrower framing could reasonably score 7–8/10 on novelty; the "build the whole platform" framing as currently written scores lower because most of the platform already exists commercially.

5. Research Gaps in Existing Systems and Papers

- **Heavy dependence on labeled data:** still true for most detection/action-recognition components (YOLO, VideoMAE fine-tuning, PPE datasets); your self-supervised VL-JEPA backbone is a legitimate partial answer to this gap.
- **Weak temporal reasoning / no long-term memory:** widely cited limitation in the anomaly-detection survey literature (e.g., need for "memory-driven zero-shot" methods like Flashback, 2025) — an opening for a memory-augmented contribution.
- **Poor cross-domain generalization:** VLM-based anomaly detectors trained/tuned on UCF-Crime/XD-Violence-style datasets often fail to generalize to industrial/factory settings, which use different cameras, layouts, and event vocabularies — a real gap your industrial focus could exploit if you build or curate a small industrial-safety benchmark.
- **Explainability:** addressed partially by VERA-style "verbalized learning" but still an open problem — small-LLM report generation is directly relevant here if you evaluate faithfulness, not just fluency.
- **High computational cost of large VLMs/LLMs:** the entire "small LLM" framing (Phi-3 Mini, SmoLLM, Qwen2.5-3B) is a legitimate, current research direction (edge-deployable multimodal reasoning) — this is one of your stronger angles.
- **Privacy concerns:** actively discussed in industry (Protex AI markets "on-premises, non-identifiable processing"); your face-blurring step is a reasonable but not novel mitigation — federated learning / on-device processing would be a stronger research angle if you wanted to lean into this gap.
- **No standardized evaluation for "surveillance report quality":** unlike detection (mAP) or anomaly detection (AUC), there is no widely agreed metric for judging whether an LLM-written shift report is accurate, complete, and non-hallucinatory. **This is arguably the single best original research contribution available to you** — proposing and validating an evaluation protocol/benchmark for LLM-generated surveillance reports would be publishable on its own.

6. How to Improve the Project — Ranked Recommendations

Idea	Research novelty	Industry relevance	B.Tech feasibility	Publication potential	Resume value
Faithfulness/hallucination evaluation benchmark for LLM-generated surveillance reports (structured events → report → automatic + human faithfulness scoring)	High (little prior standardization)	High (every vendor above needs this)	High — mostly evaluation design + a moderate-size labeled event set	High — clean, self-contained contribution, workshop-track feasible	High
VL-JEPA vs. CLIP/VLM embeddings for event-similarity retrieval — controlled ablation	High (VL-JEPA is unstudied outside its own paper)	Medium-high	Medium — needs the VL-JEPA checkpoint/code to actually run, some engineering risk given the paper's youth	High if it works, but carries technical risk (immature codebase)	High
Memory-augmented long-horizon event reasoning (e.g., "has this person loitered near Gate 2 three times this week?")	High	High	Medium — requires designing a memory module, not just retrieval	Medium-high	High
RAG over the structured event DB + vector store for natural-language querying	Medium (VideoRAG precedent exists) but domain-specific application is fresh	High	High — very achievable with existing RAG tooling	Medium	High
Knowledge-graph construction from events (people, machines, zones, time) for causal/predictive alerting	High (MissionGNN precedent, but not applied to structured industrial timelines)	High	Medium	Medium-high	High
Edge/on-device deployment of the small LLM + JEPA pipeline (quantization, latency/throughput study)	Medium	High (matches real deployment constraints)	High	Medium	High
Open-vocabulary / zero-shot event detection (detect event types not seen at training time)	High	Medium-high	Medium-low (harder to execute well in the time available)	Medium-high	Medium-high
Cross-camera event fusion / re-identification	Medium (active, crowded research area — see Section 2)	High	Medium-low (Re-ID is a hard, mature sub-field; easy to under-deliver)	Medium	Medium

Idea	Research novelty	Industry relevance	B.Tech feasibility	Publication potential	Resume value
Predictive maintenance from historical event sequences	Medium	High	Low-medium (needs substantial historical labeled sequence data you likely won't have)	Low-medium	Medium
Federated learning for multi-site privacy	High academically	Medium (real deployments rarely do full FL yet)	Low for a solo/small capstone team	Medium	Medium
Full explainable-AI layer (attention/saliency overlays explaining each detection)	Medium	Medium-high	Medium	Medium	Medium

Recommendation: pick **one** primary contribution (the faithfulness-benchmark idea or the VL-JEPA-vs-CLIP ablation are the strongest) and **one** secondary feature (RAG-based NL search, or a lightweight knowledge graph) rather than attempting all 20+ events and every advanced feature listed in the original brief. Breadth across 20 event types is a systems-engineering achievement, not a research one — reviewers and interviewers will care more about a sharp, evaluated claim than a long feature list.

7. Feasibility Analysis for a B.Tech Capstone

Hardware: A single modern GPU with ≥16 GB VRAM (RTX 4070 Ti/4080 class or a cloud A10/L4 instance) is realistically sufficient if you fine-tune/adapt rather than pretrain from scratch — VL-JEPA and V-JEPA-class backbones are meant to be used as frozen or lightly fine-tuned feature extractors, not retrained end-to-end. Small LLMs (Phi-3 Mini ~3.8B, Qwen2.5-3B, SmoLLM) run comfortably at 4-bit/8-bit quantization on 8–16 GB VRAM, or even on CPU with acceptable latency for a report-generation (not real-time chat) use case.

Datasets: No public dataset matches your full industrial event taxonomy (worker+vehicle+machine+safety events together). Realistic options: (1) combine existing public datasets per sub-task — UCF-Crime/XD-Violence (anomaly), SH17 (PPE), a fall-detection dataset (e.g., the YOLOv8-based industrial fall dataset referenced above), Facility-ReID (industrial multi-camera Re-ID) — and build your own small labeled set for machine/conveyor-specific events, since none of these exist publicly at meaningful scale. Budget significant time for this;

dataset assembly, not model architecture, is usually the actual bottleneck in projects like this.

Timeline (typical 2-semester B.Tech capstone, team of 3–4):

- Months 1–2: literature review, dataset assembly/curation, environment setup, VL-JEPA feasibility spike (does it run, can you extract usable embeddings on your hardware?)
- Months 2–4: detection/tracking pipeline, event schema + database, baseline report generation with the small LLM
- Months 4–6: your chosen "sharp" research contribution (evaluation benchmark, ablation, RAG layer, etc.) + dashboard
- Months 6–8: evaluation, ablations, writing, poster/demo, paper draft if pursuing publication

Team size: 3–4 is realistic for the scoped-down version; the full original brief (20+ event types, predictive alerts, federated learning, multi-camera fusion, knowledge graphs, edge deployment, dashboard, PDF/CSV export) genuinely reads like a 6–12 month product roadmap for a funded team, not one capstone team's deliverable — expect to cut ~60–70% of the listed advanced features.

Software: All named stack components (React/TypeScript, FastAPI, OpenCV/FFmpeg, PostgreSQL, pgvector/FAISS, Redis, Docker) are appropriate, mature, and low-risk choices.

Deployment/scalability challenges: multi-camera, always-on video pipelines are I/O- and storage-heavy even before modeling; plan frame-sampling and storage-retention policy early. GPU inference cost for 1–5 FPS across many cameras is the main real-world scaling constraint (this is exactly why NVIDIA's VSS and Verkada emphasize edge/on-camera processing).

Ethical/privacy considerations: face blurring is good practice but not sufficient on its own — document a clear data-retention policy, consent/notice framework for monitored premises, and be explicit in your report about bias/fairness risks in person detection and Re-ID, which are well-documented failure modes in the surveillance-AI literature.

8. Publication Potential

Realistically **not** a top-tier venue (CVPR/ICCV/NeurIPS) submission as scoped — those require large-scale benchmarks, extensive baselines, and novel theory beyond what a capstone timeline allows. Realistic and credible targets:

- Workshops** at CVPR/ICCV/WACV on video anomaly detection, embodied/edge AI, or applied vision (many of the papers cited above are workshop or specialized-venue papers, e.g., MDPI Sensors, WACV).
- Applied/industry tracks:** IEEE Access, MDPI Sensors/Electronics/Sustainability (multiple cited construction-safety papers appeared here), or a regional/national student conference.
- arXiv preprint + GitHub** is a very achievable and resume-valuable minimum bar regardless of formal acceptance.

What you'd need to add for a credible submission:

- A named, reproducible benchmark or dataset slice (even small) with clear splits.
- Comparison against at least 2–3 baselines (e.g., a CLIP-embedding retrieval baseline, a supervised-only detection pipeline without VL-JEPA, and/or a larger cloud VLM like the ones NVIDIA VSS uses) — this is what turns "we built a system" into "we show X outperforms Y under Z conditions."
- Standard metrics: mAP/F1 for detection; AUC/precision-recall for anomaly-style events; BLEU/ROUGE alone are weak for report quality — pair them with a faithfulness/factual-consistency metric and, ideally, a small human evaluation.
- An ablation study isolating the effect of the self-supervised backbone (with/without VL-JEPA vs. a supervised alternative) — this is the paper's core empirical claim.

9. Final Evaluation

Dimension	Score (1–10)	Rationale
Overall originality	6	Strong at the specific-combination level (VL-JEPA + small LLM + industrial CCTV); weak at the system-level idea, which closely parallels shipping commercial products
Research depth (as currently scoped)	5	Broad feature list reads as engineering scope, not a research question; needs one sharpened contribution
Industry relevance	9	Matches an active, fast-growing, well-funded market (Verkada, NVIDIA, BriefCam/Milestone, Protex AI, viAct all compete here right now)
Resume value	8	Full-stack, multimodal AI, real deployment constraints — strong signal to employers regardless of publication outcome
Startup potential	4	Market is already crowded with well-funded incumbents (Verkada alone raised at multi-billion valuation); a startup would need a very sharp wedge (e.g., cheap on-prem industrial safety for SMEs) rather than "general CCTV AI"
Publication potential	6	Workshop/applied-venue realistic with the right ablation and evaluation design; top-tier unlikely at capstone scale
Technical difficulty	7	Genuinely hard due to VL-JEPA's immaturity, multi-modal pipeline integration, and dataset assembly burden — moderate-high for a B.Tech team

Top 5 Recommended Modifications to Maximize Novelty and Impact

- Narrow the research question.** Reframe from "build a full CCTV platform" to "does a JEPA-style self-supervised video-language backbone (VL-JEPA) improve efficiency and/or generalization over CLIP/VLM-based pipelines for industrial-safety event understanding and report generation?" — this single reframing is what separates a capstone from a hackathon demo.
- Build (even a small) faithfulness/hallucination evaluation protocol for the generated reports.** This is the least-crowded, most publishable piece of the entire proposal, and it's achievable with modest scope (a few hundred labeled report/ground-truth pairs plus human eval).
- Cut the event taxonomy to ~6–8 well-instrumented events** (e.g., PPE missing, fall, restricted-area entry, machine stop, fire/smoke, unauthorized vehicle) rather than 20+, and get those working reliably with real evaluation numbers — depth beats breadth for both research credibility and demo reliability.
- Add one comparison baseline you can run yourself:** a CLIP-embedding or off-the-shelf VLM (e.g., a small open Qwen2.5-VL) pipeline for the same task, so your VL-JEPA choice is empirically justified rather than asserted.
- Position the industrial-safety vertical explicitly against Protex AI / viAct,** not against general security incumbents like Verkada/Hikvision — you'll get a much more honest, defensible "gap in the market" story (privacy-preserving, self-supervised, small-LLM, low-cost, on-prem — for SMEs that can't afford \$800–\$2,500/site/month enterprise platforms) than trying to out-position Verkada or NVIDIA in general CCTV AI.